

Spatial model of Real Estate portfolio exposure - Cholesky decomposition

$$Y(x) = ay(x) + bw(x) \int_0^T Z_w(sx) ds + Z_y(Tx),$$

where $Z_y(tx)$ is a partly time-changed Brownian sheet

$$Z_y(tx) = y \exp(y_1 x_1 + y_2 x_2) W_t^{v_y},$$

with the time changes

$$v_y^2(x_1) = \exp(2y_1 x_1), \quad v_y^3(x_2) = \exp(2y_2 x_2),$$

- deterministic component $ay(x)$, the integral of temperature deviations from the trend $\int_0^T Z_w(sx) ds$ weighed by a factor $bw(x)$, and a noise term $Z_y(Tx)$.

$$\text{Cov}[Z_y(tx)Z_y(sy)] = 2 \exp(y_1(t-s) + y_2(x_2 - y_2)).$$

$x_f = (x_1, \dots, x_d)$ are geographical positions. If the correlation matrix of $Z_y(tx_f)$ has the Cholesky decomposition

$$\text{corr}(Z_y(tx_f), Z_y(tx_k)) = \exp(y_1(x_{f1} - x_{k1}) + y_2(x_{f2} - x_{k2})),$$

then the vector $Z_f y(t)$ has the following representation

$$Z_f y(t) = Z_f y(0) + y W_t^y,$$

where W_t^y is a vector of independent standard Brownian motions. $w \in \mathbb{R}^{p+d}$ is the Cholesky decomposition of the spatial correlation matrix of $Z_w(tx)$ at locations $x_1, \dots, x_p$. Diagonal matrices

$$b_w = \text{diag}(b_w(x_f)), \quad h_w = \text{diag}(w(x_h)), \quad f_w = \text{diag}(w(x_f)),$$

RE thus follows a multivariate normal distribution with moments -

$$E[Y|F_t] = ay + bw \int_0^T Z_f w(s) ds + bw(1 - e^{w(T-t)})Z_f w(t) + Z_f y(t),$$

and

$$\text{Var}[Y|F_t] = \int_0^t (1 - e^{w(T-s)})^2 ds + bw f_w (h_w^T + f_w^T) f_w b_w + 2y(T-t)y^T y.$$

Agent Portfolios

Property Managers - portfolio holdings taken as of 2024 Financial Year, and weighted average extrapolated based on portfolio MV% (market value). The rent valuation time series is proxied by the valuation of nearest estate referenced from the Centaline Big Data transactions database.

The intersection criterion considered is a **ball tree algorithm**, which is a spatial indexing method utilizing **Harviside distance minimization** for spatial proximity. Inputs required are geographical coordinates (latitude and longitude) to create a hyper sphere.

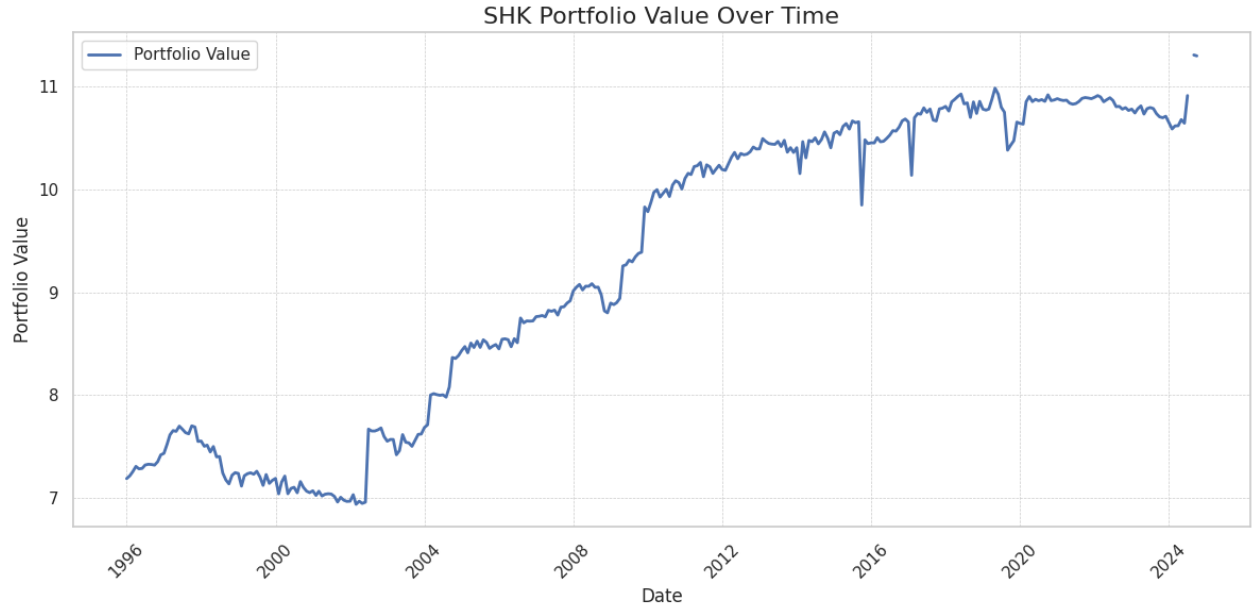


Figure 1: SHK Properties Manager - Time Series Hong Kong Portfolio's Rent Valuation. Portfolio based on weighted average of estates nearest to properties currently in SHK's portfolio as of 2024

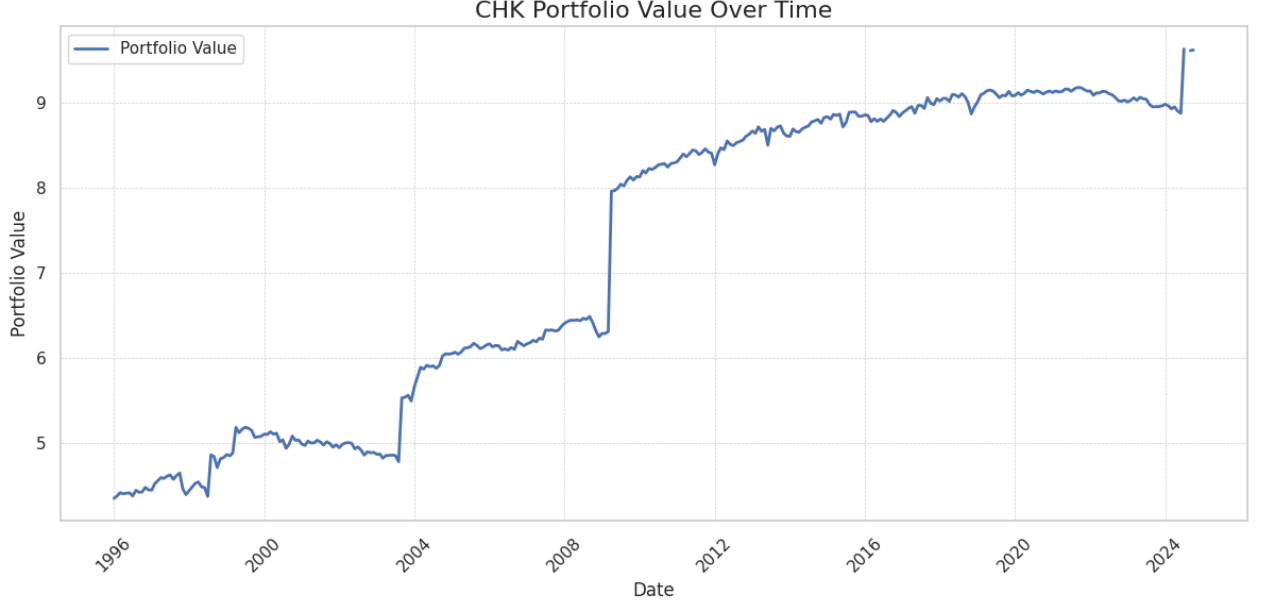


Figure 2: CHK Properties Manager - Time Series Hong Kong Portfolio's Rent Valuation.

Spatial and Temporal Effects Regression

The relationship between housing prices and climatic, temporal, and district characteristics can be expressed as follows:

$$\begin{aligned} \ln(\text{Portfolio rent valuation}_{i,t}) = & \beta_1 \ln(\text{temperature}_{i,t}) + \beta_2 \ln(\text{temperature}_{i,t}) \times \ln(\text{rainfall}_t) + \\ & \beta_3 (\text{number of severe typhoons}_t) \times (\text{number of coastal risk spots}_i) + \\ & \beta_4 \ln(\text{rainfall}_t) \times (\text{number of flooding blackspots}_{i,t}) \times (\text{climate news index}_t) \\ & + \delta(\text{other control variables})_{i,t} + \text{year effect} + \text{district fixed effect} + \epsilon_{i,t}. \end{aligned}$$

District Characteristics (Explaining district fixed effect) The district fixed effect accounts for cross-sectional variations in the following district specific characteristics:

- % built-up
- % vegetated
- % bare soil
- % rock
- % water
- % dense/high vegetation

- % sparse/low vegetation
- Building fraction (%)
- Road/impervious cover fraction (%)
- Average Sky View Factor (SVF)
- Average building height (m)
- Standard deviation of building height (m)